

the features most wanted by prospective car buyers. The study also suggested that fuel consumption, cost-performance ratio, safety, price, design and comfort are considered to be the major selling points.

This is good news for the automotive electronics sector, which has been hard at work developing power train and chassis improvements designed to reduce fuel consumption and improve long-term cost-performance ratio levels. Factors like safety, comfort, telematics and in-car infotainment are also some of the current major industry themes. However, at the same time, it is important to note that Indian consumers who drive medium- or low-cost vehicles are interested in having the latest features in their vehicles that are also price sensitive in nature.

According to Pai, "Today, car makers are competing to bring in advanced features, connectivity, entertainment, advanced safety and improved fuel efficiency to their products to attract potential buyers. To address this ever-increasing list of consumer needs, more and more technological advancements are, and will keep, happening, and automotive electronics will play a pivotal role in satisfying all user requirements. This paves the way for complex electronic systems. Therefore providing cost-competitive electronic solutions is a challenge and an opportunity for Indian automotive companies."

Safer, smarter, greener options

According to an iSuppli Corp. report, technological development in engine management and incorporation of safety systems in cars are expected to be the key growth drivers for the Indian automotive electronics market in the next five years as it attains a compound annual growth rate (CAGR) of 21.8 per cent.

Electronics plays a key role in improving road safety. Safety applications in India are currently policy-driven, possibly due to the cost sensi-

Electronic consumption by the automotive segment

Power train and chassis

- Diverse electric motors
- Electrical (turbo) charging
- New igniters and advanced sensors
- Automated manual transmission
- e-braking, e-steering, e-suspension
- Electronic stability control
- Start-stop system

Safety

- Rear cameras
- Lane departure warning
- Adaptive cruise control
- Blind-spot detection
- Map-supported adaptive front lighting
- Tire-pressure monitoring system
- Curve warning
- Speed alert
- Collision warning/avoidance
- Usage data transmission

Telematics and infotainment

- Fixed and portable navigation devices
- Multimedia features
- Bluetooth
- eCall
- Voice based, haptic and human-machine interface (HMI) systems and touch
- Real-time traffic information
- Parking assistants
- Social media
- Wi-Fi hotspot, smartphone interface

Comfort and convenience

- Adaptive front lighting
- Steering wheel buttons
- Central/head-up displays
- Voice-control systems
- Xenon headlights
- LED/ambient lighting
- Touchscreens
- Remote control
- Well-being assistants

Source: Frost & Sullivan Report, 2014

tivity of the Indian market. Car safety is set to receive a serious boost in the country with the government planning to make ABS, airbags and other safety equipment compulsory on all passenger cars sold in India. As per the proposal, India will have its own vehicle safety standard called India New Car Assessment Programme (NCAP), which is similar to United Nations' NCAP.

This would present a serious challenge for ultra-low-cost and two-wheeler segments as safety features found typically in higher-segment cars are expensive. This will also, no doubt, force automotive innovation in yet another direction, as car makers and suppliers attempt to meet emission and safety standards at a low cost.

According to Merli, increasing electronics content in the two-wheeler segment is likely to fuel the demand for optimised semiconductor solutions suitable for cost-sensitive markets.

Merli also pointed towards the huge scope that exists in the com-

mercial vehicles space. In India, safety and emissions in commercial vehicles are a huge concern for regulatory authorities. This might result in setting new mandates in this segment. For example, ABS is now mandatory for commercial vehicles, which may be followed by regulations in tire pressure-monitoring systems (TPMSes). And eventually, all these will result in increasing the electronics content in commercial vehicles, too.

Emission reduction also poses a major challenge for the Indian commercial vehicle manufacturers, and regulations with respect to emission would be made mandatory in India soon, which would provide an impetus for growth in the Indian automotive segment, especially in power train and transmission.

In addition to ABS, incorporation of some advanced driver assistance systems (ADASes) will also help to make Indian cars safer and will definitely increase the electronics content of the cars. In this regard, Pai referred